

Numerical skills for neuroscience

Part 1 · Functions, change and rhythms

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Three questions to carry into a paper

AN INTRODUCTION FOR MSC NEUROSCIENCE

A mathematical tool

A question it helps us ask

Function $y = f(x)$

How are two quantities related?

Derivative dy/dt

How quickly does something change?

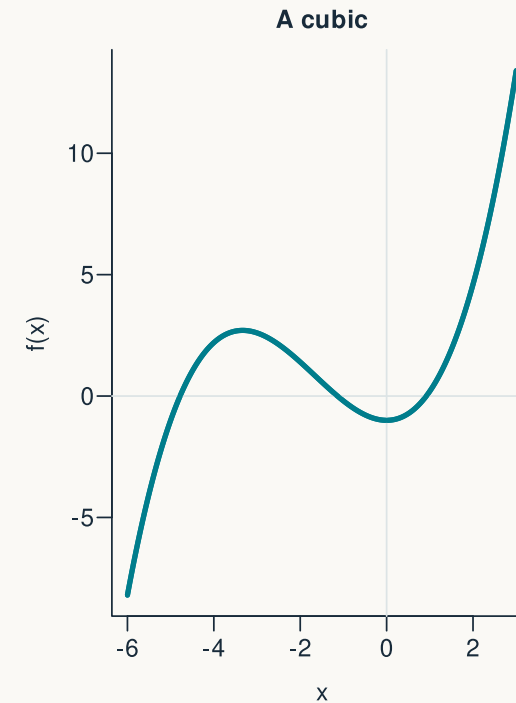
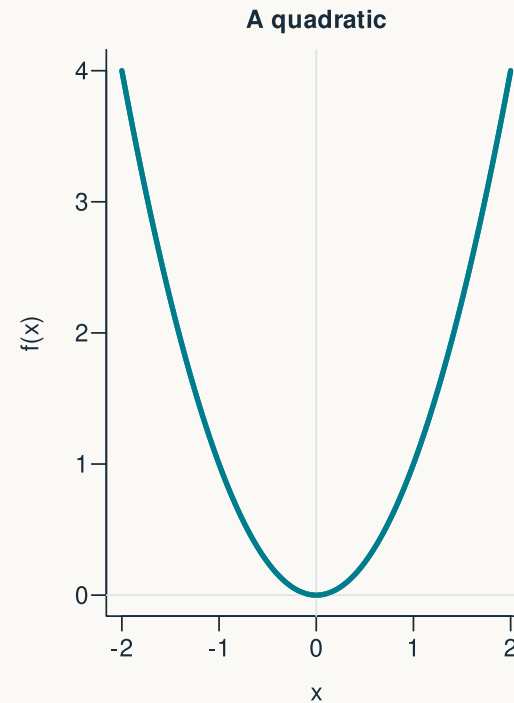
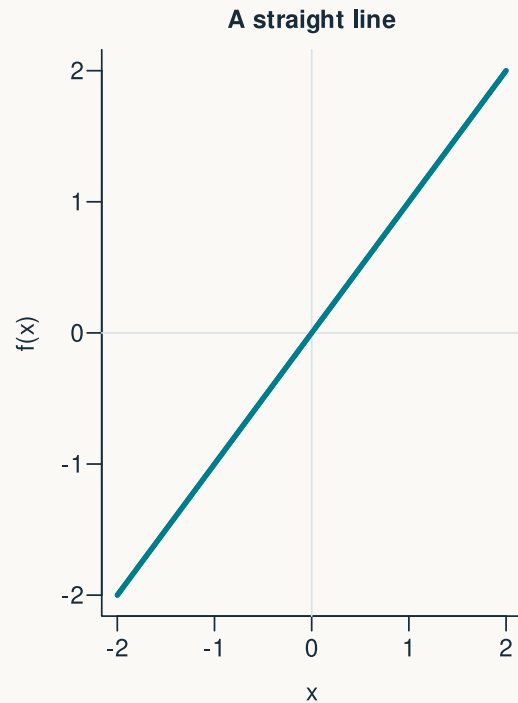
Fourier transform

Which rhythms are present?

Aim: read a graph or equation and explain what it means.

A function maps an input to an output

$$x \longrightarrow f(x) \longrightarrow y$$



$$f(x) = x$$

$$f(x) = x^2$$

$$f(x) = \frac{1}{5}x^3 + x^2 - 1$$

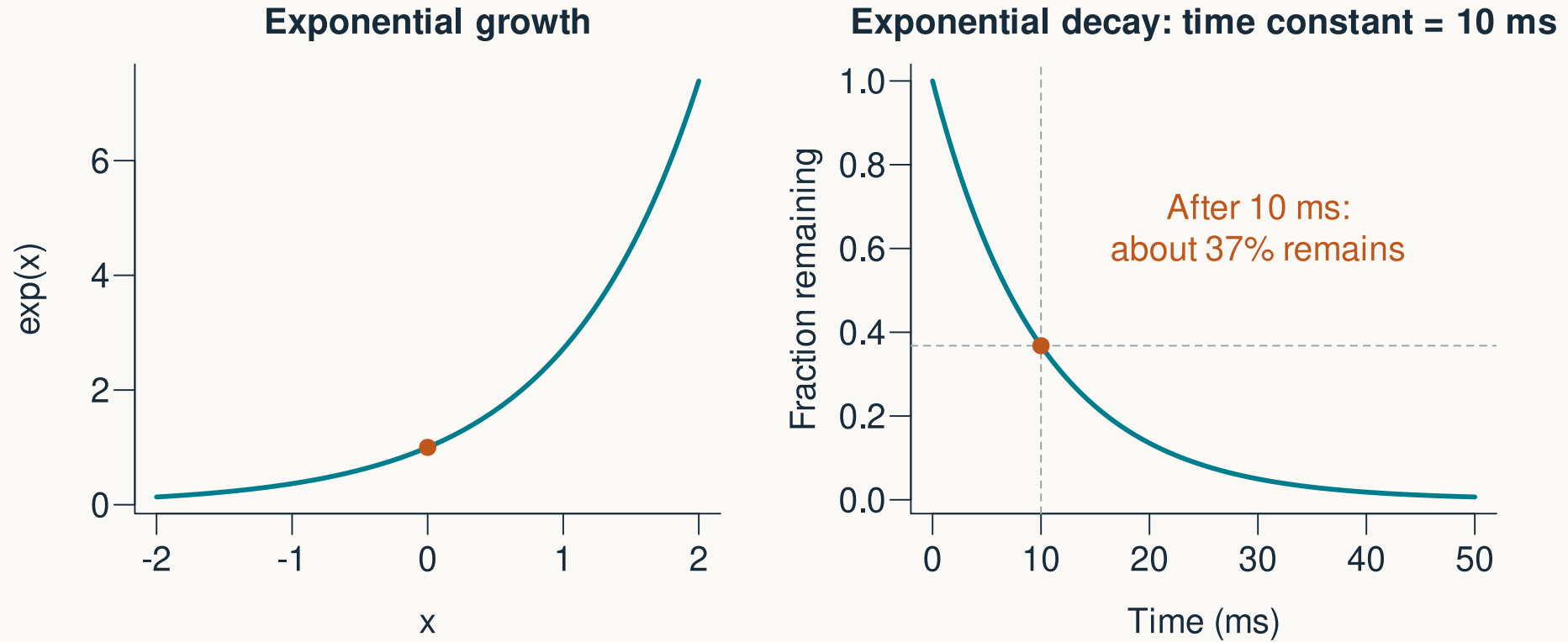
Read the symbols before doing the algebra

$$\underbrace{V(t)}_{\text{voltage at time } t} = \underbrace{V_0}_{\text{starting value}} e^{-t/\tau}$$

Symbol	Read it as
t	Input: elapsed time
$V(t)$	Output: voltage relative to baseline
V_0, τ	Parameters: values that set the curve's shape

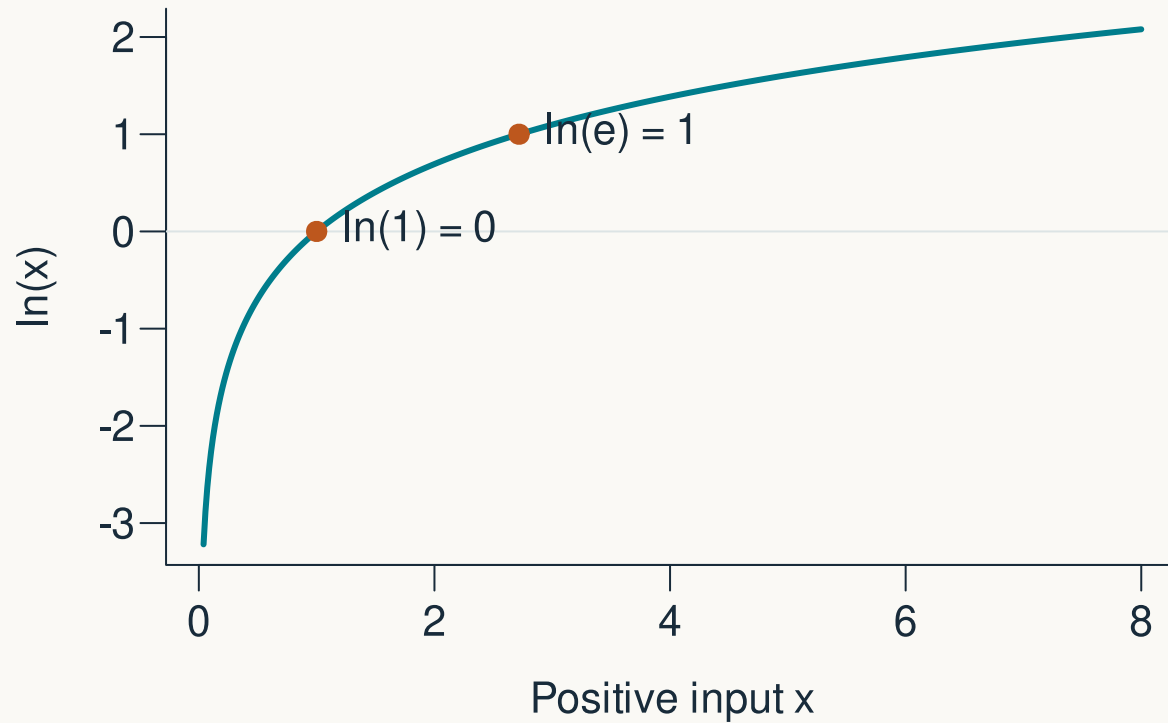
Check the units: t and τ must use the same time unit.

Exponentials describe growth and decay



$$\exp(x) = e^x \quad e \approx 2.718 \quad V(t) = V_0 e^{-t/\tau}$$

A logarithm reverses an exponential



$$\ln(e^x) = x$$

$$\ln(ab) = \ln(a) + \ln(b)$$

For real logarithms: **positive inputs**.

In R: `exp(x)` and `log(x)`.

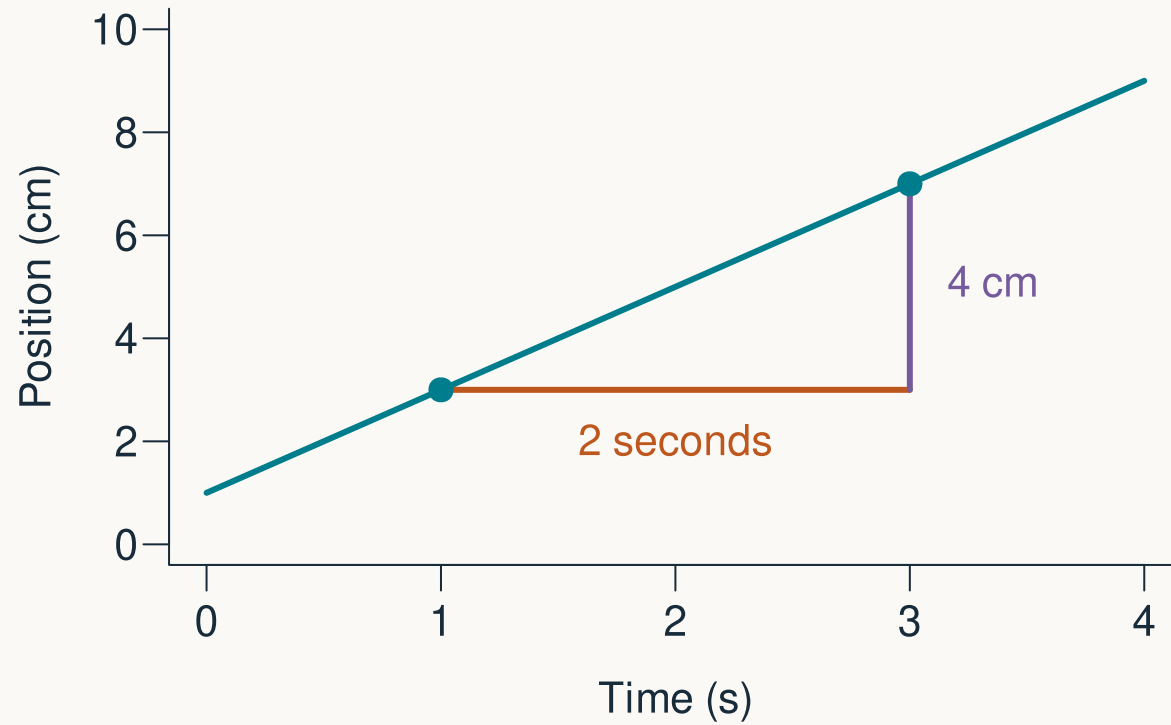
Pause: read the curve

For $V(t) = V_0 e^{-t/\tau}$, doubling τ makes the decay...

A. faster **B.** slower **C.** unchanged

B. Slower. It takes twice as long to reach the same fraction of V_0 .

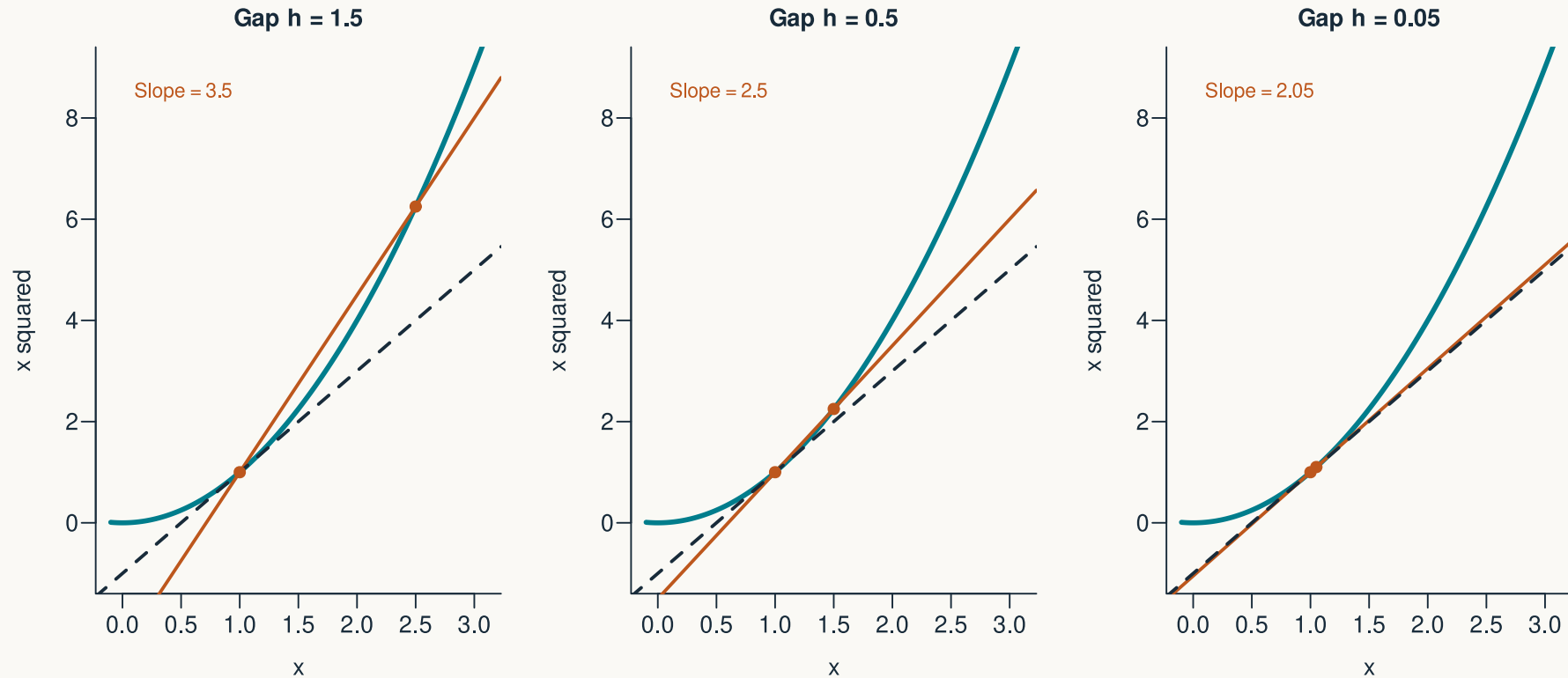
A slope is a change divided by a change



$$\frac{\Delta y}{\Delta t} = \frac{4 \text{ cm}}{2 \text{ s}} = 2 \text{ cm/s}$$

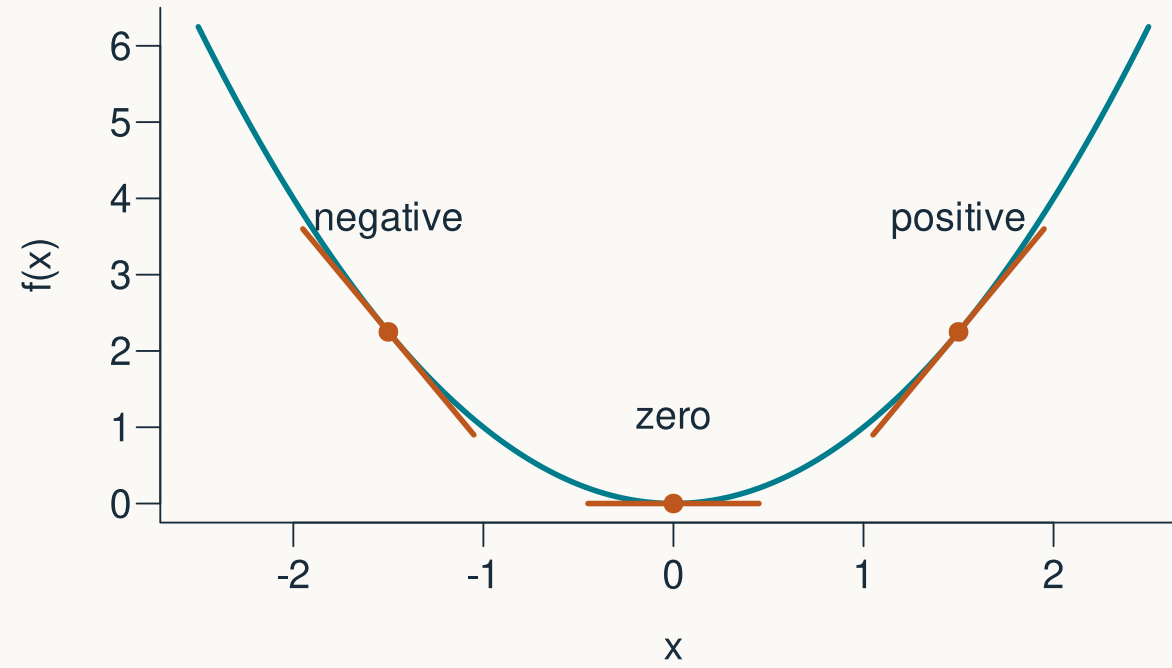
Δ means **change in**.

On a curve, the slope depends on where you are



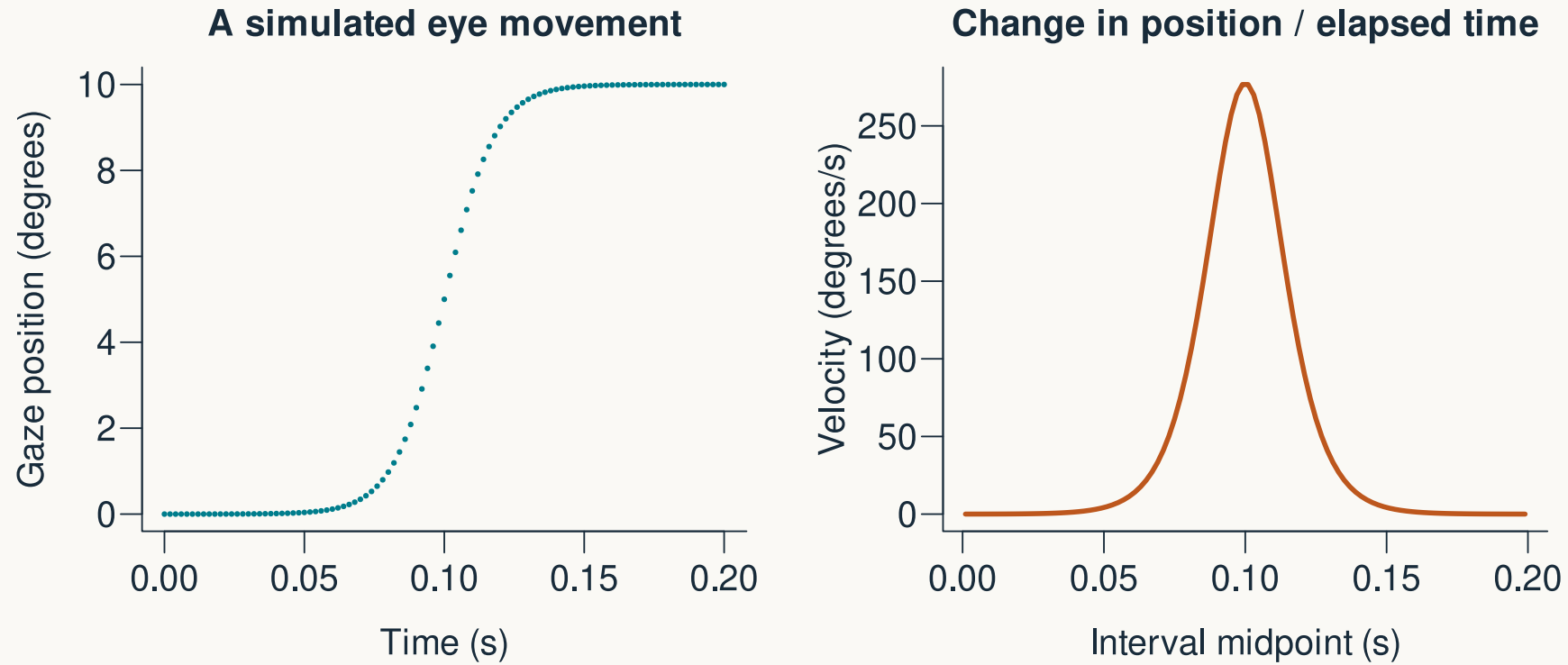
$$f'(x) = \frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

A derivative tells us the local direction of change



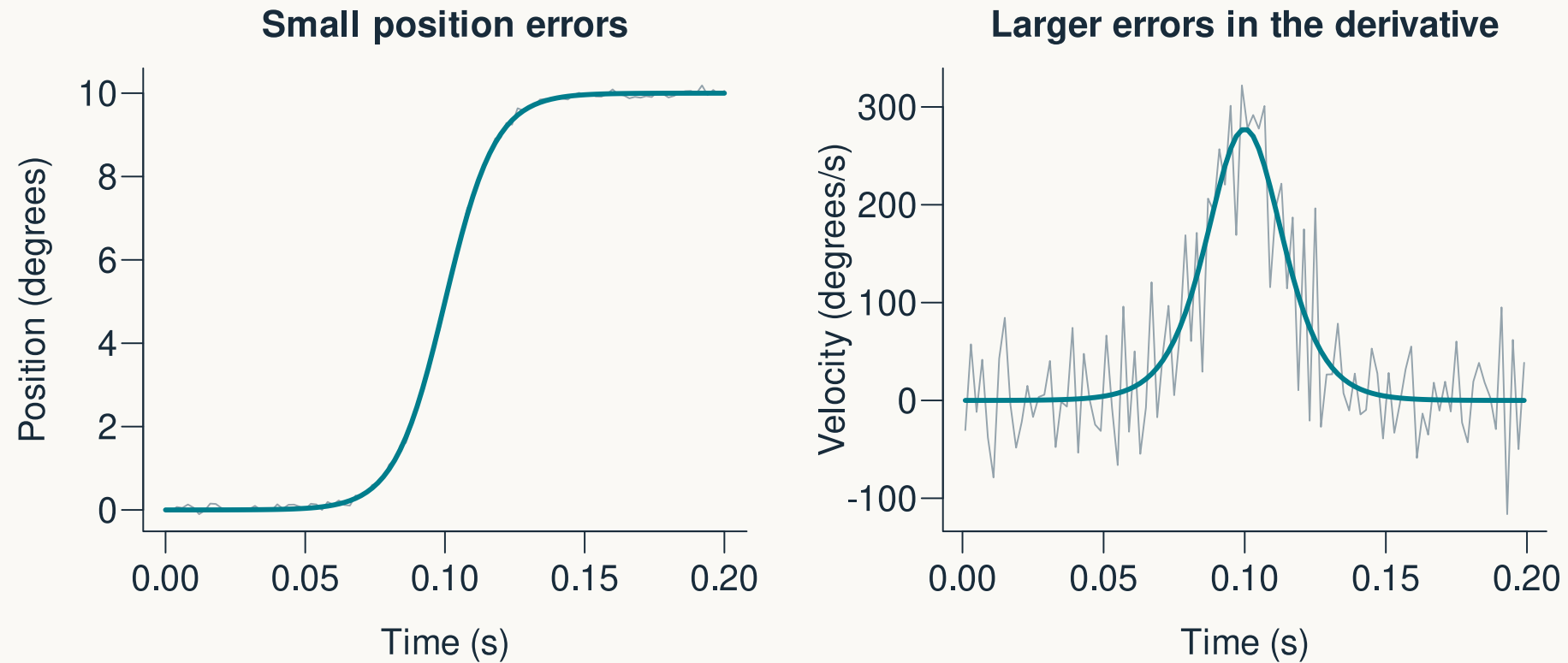
$$f(x) = x^2 \quad \Rightarrow \quad f'(x) = 2x$$

From gaze position to gaze velocity



$$v_{i+1/2} \approx \frac{x_{i+1} - x_i}{t_{i+1} - t_i}$$

Small measurement errors can become large derivative errors



Differencing also measures changes in the noise.

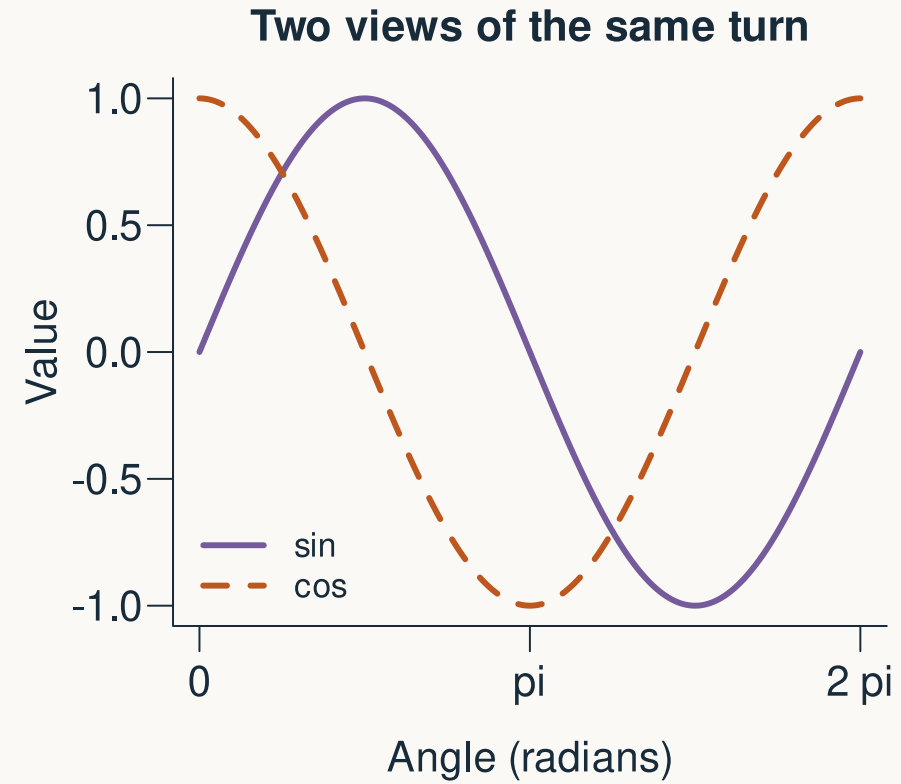
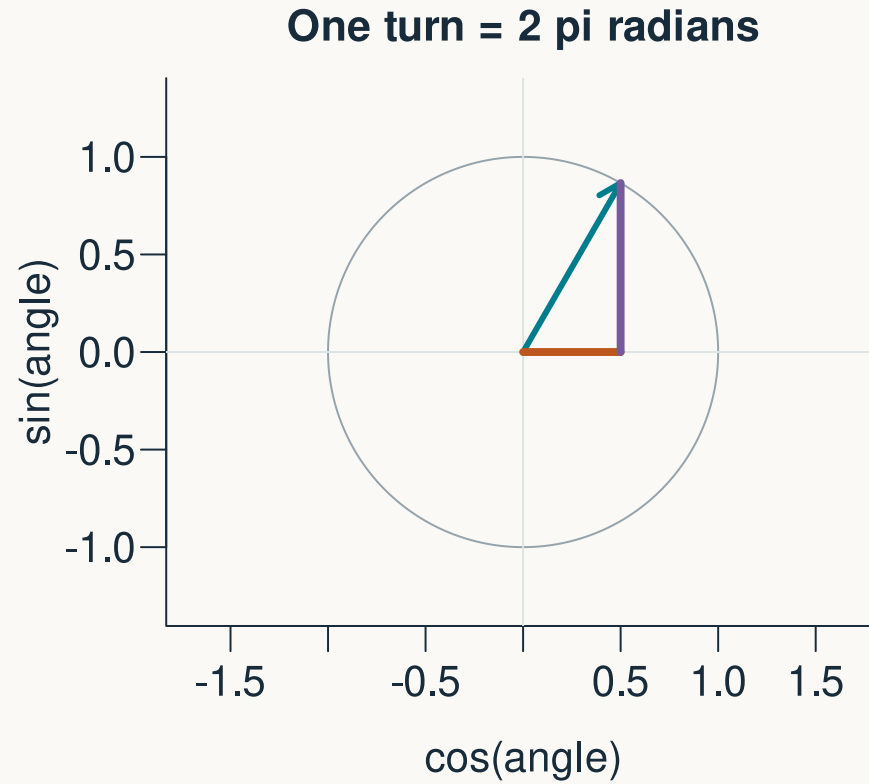
Pause: keep track of the units

Gaze moves by 0.6° between samples separated by **2 ms**.

What is its average velocity over this interval, in degrees per second?

$$\frac{0.6^\circ}{0.002 \text{ s}} = 300^\circ/\text{s}$$

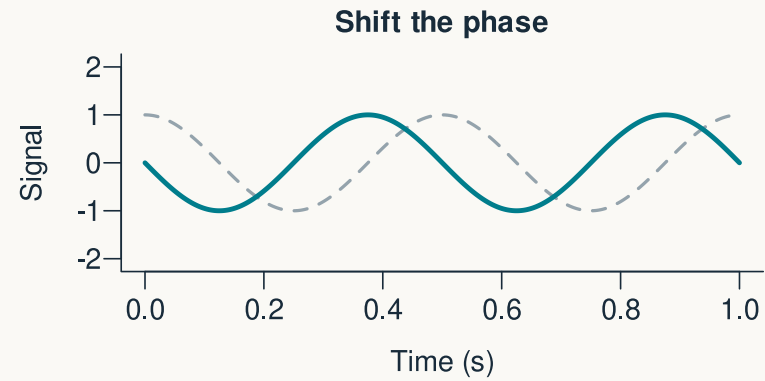
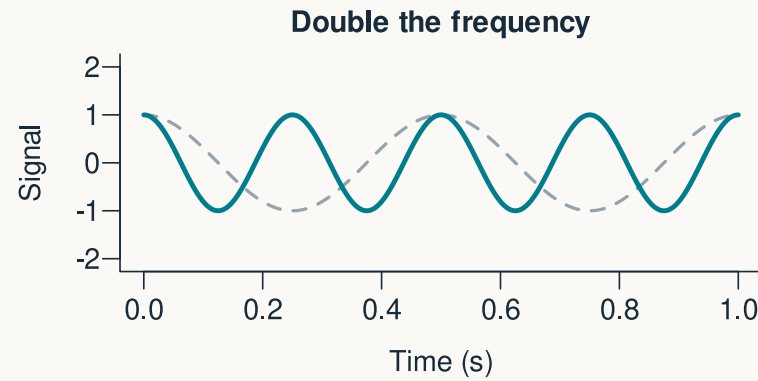
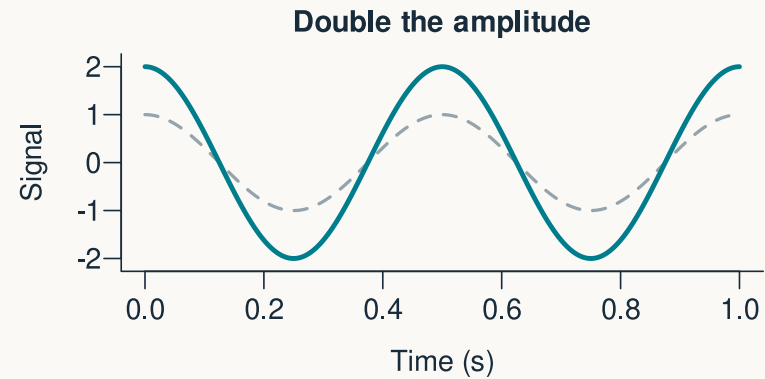
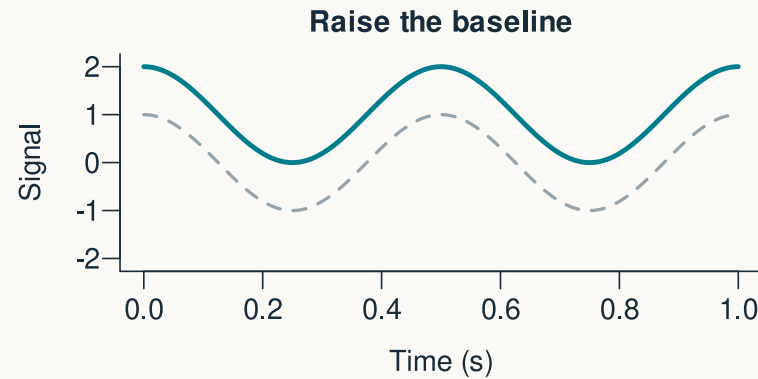
Sine and cosine turn angles into oscillations



$$360^\circ = 2\pi \text{ radians}$$

Four quantities set a simple rhythm

$$y(t) = b + A \cos(2\pi f t + \phi)$$



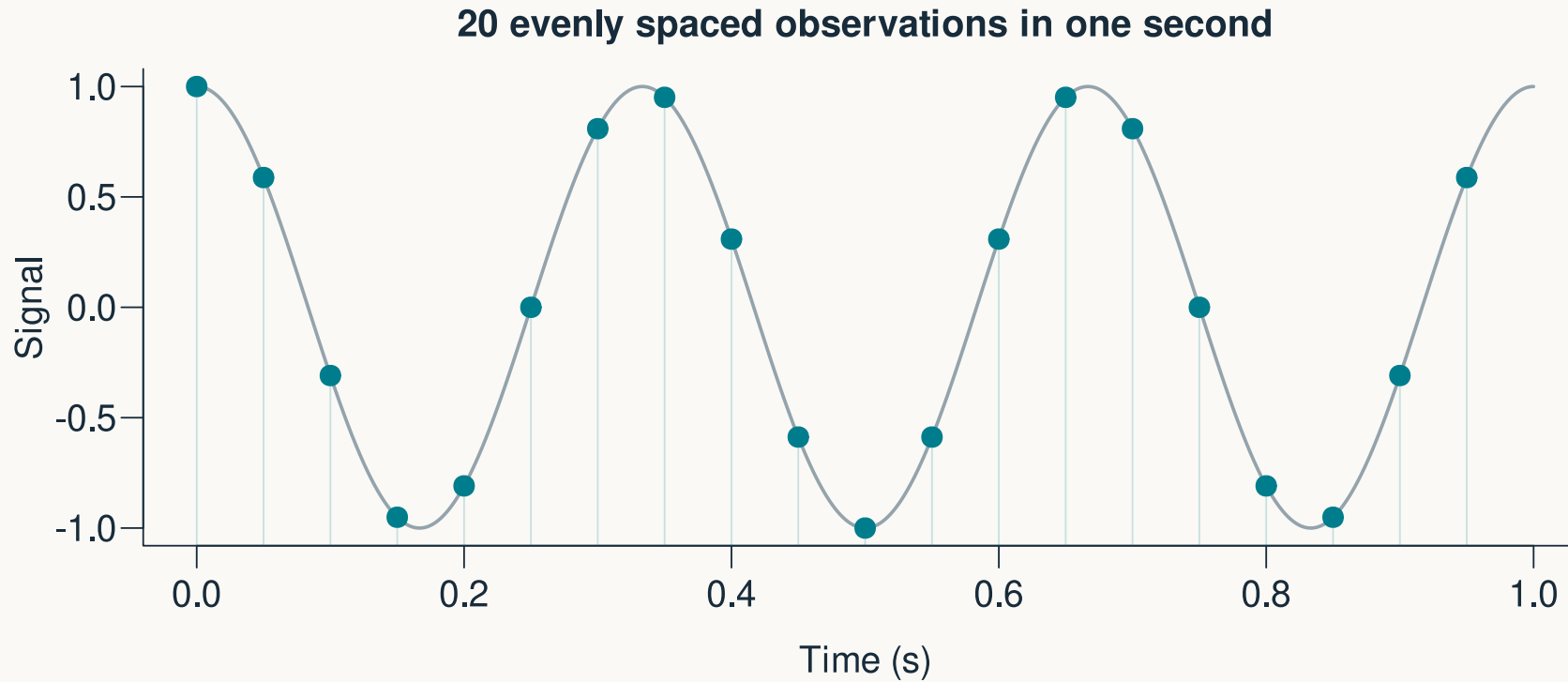
Period and frequency are reciprocals

$$f = \frac{1}{T}$$

One cycle takes...	Frequency	Units
0.1 seconds	10	Hz = cycles/second
1 day	1	cycle/day
28 days	$1/28 \approx 0.036$	cycles/day

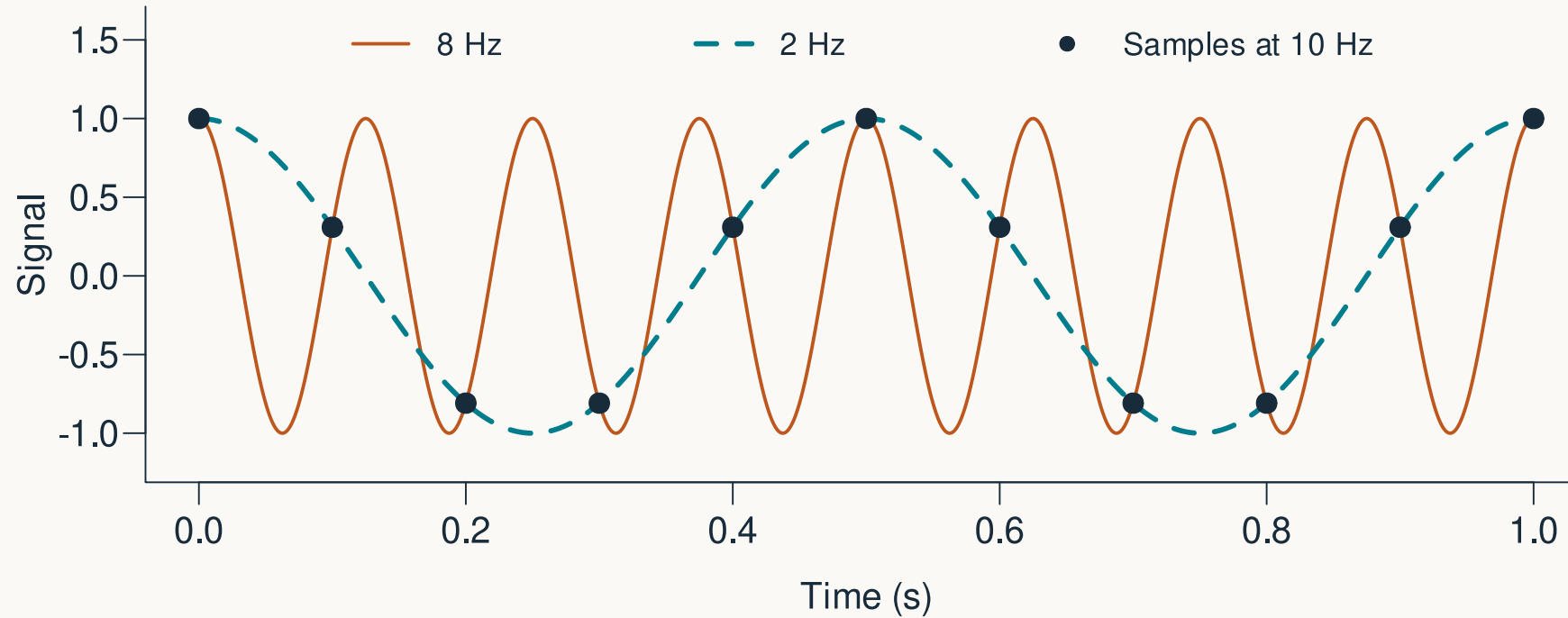
If the period halves, what happens to the frequency?

A recording is a sequence of samples



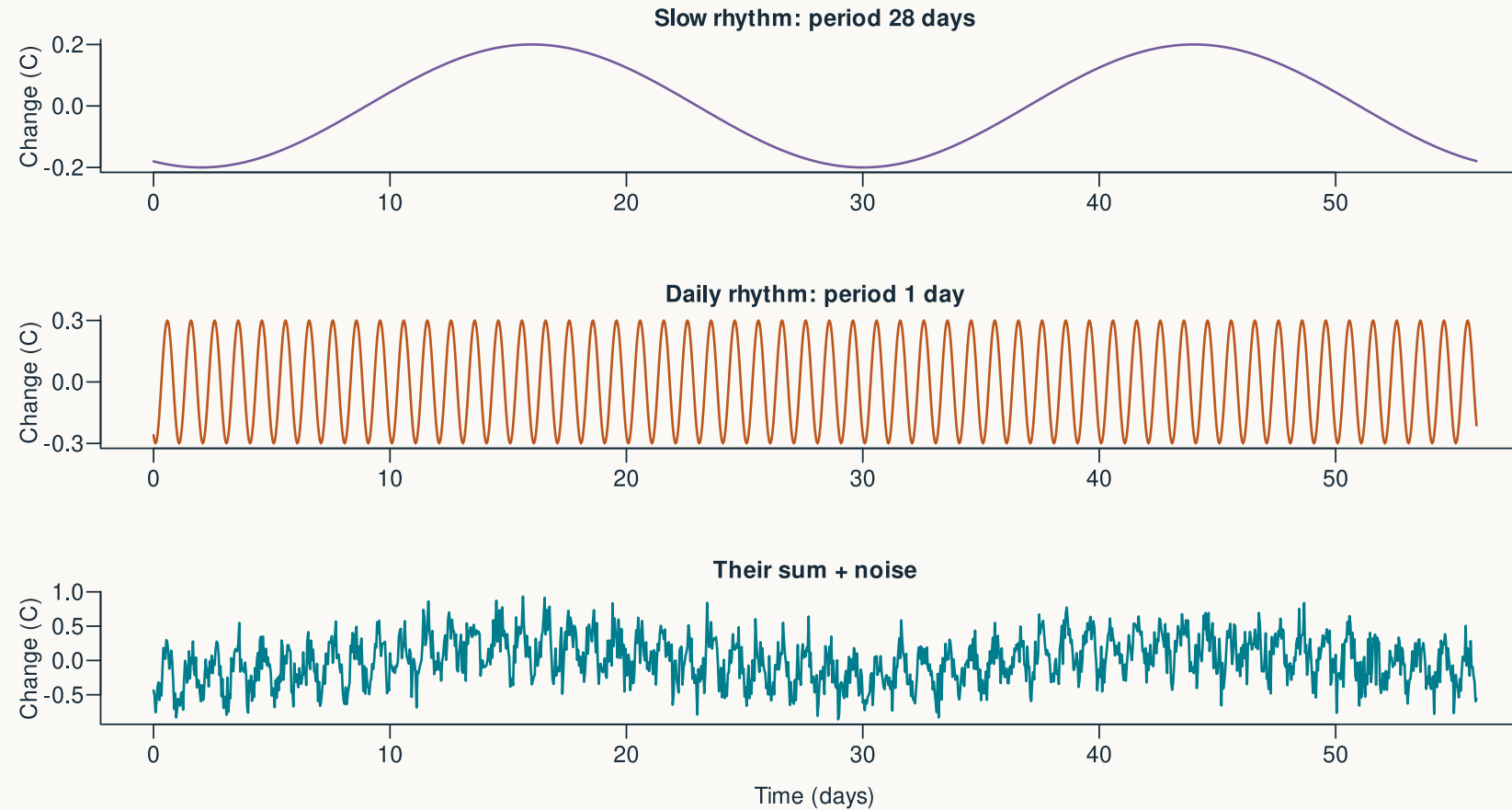
$$f_s = \frac{1}{\Delta t} \quad t_i = \frac{i}{f_s}, \quad i = 0, \dots, N - 1$$

Sampling too slowly can disguise a rhythm



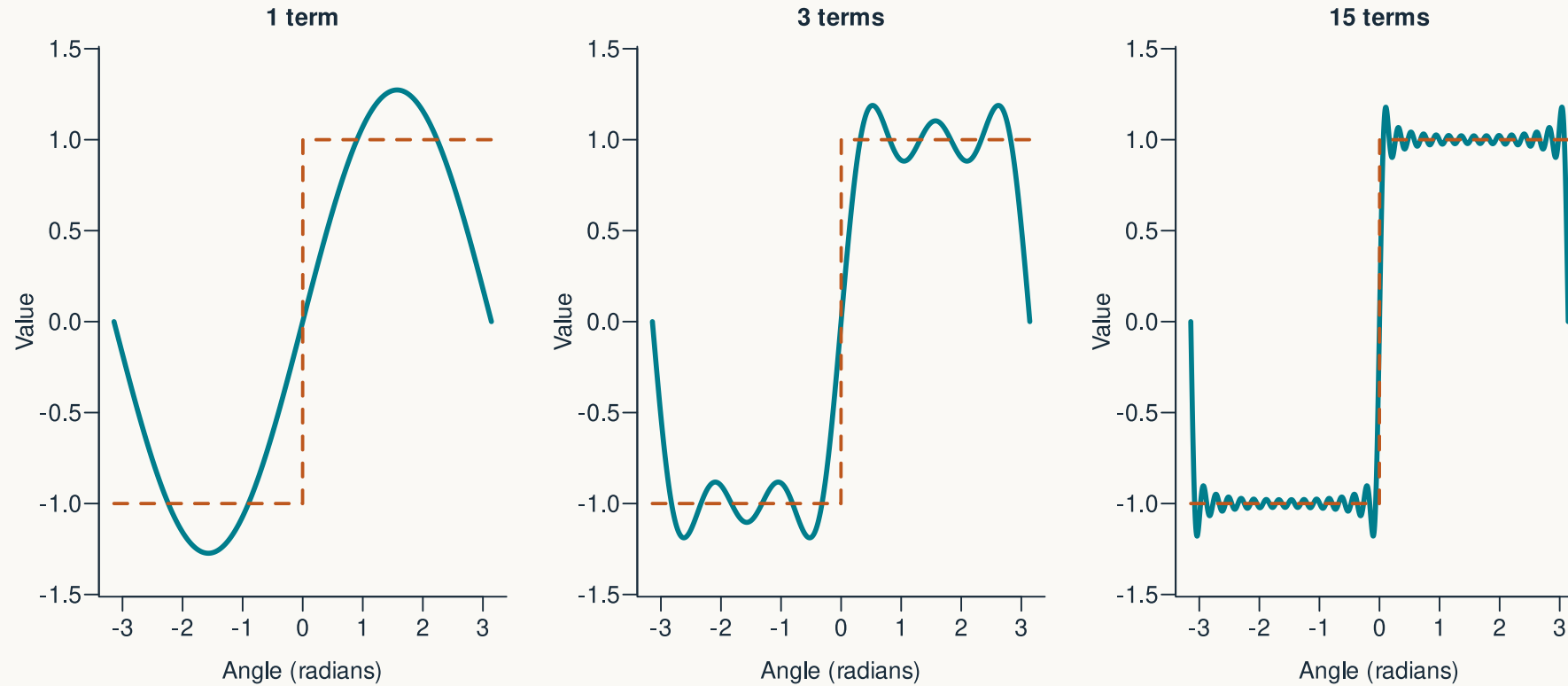
For a band-limited signal: $f_s > 2f_{\max}$, with room for anti-alias filtering.

One signal can contain several rhythms



Synthetic temperature-like signal · 56 days · hourly samples · baseline removed here

Adding sinusoids can build more complex shapes



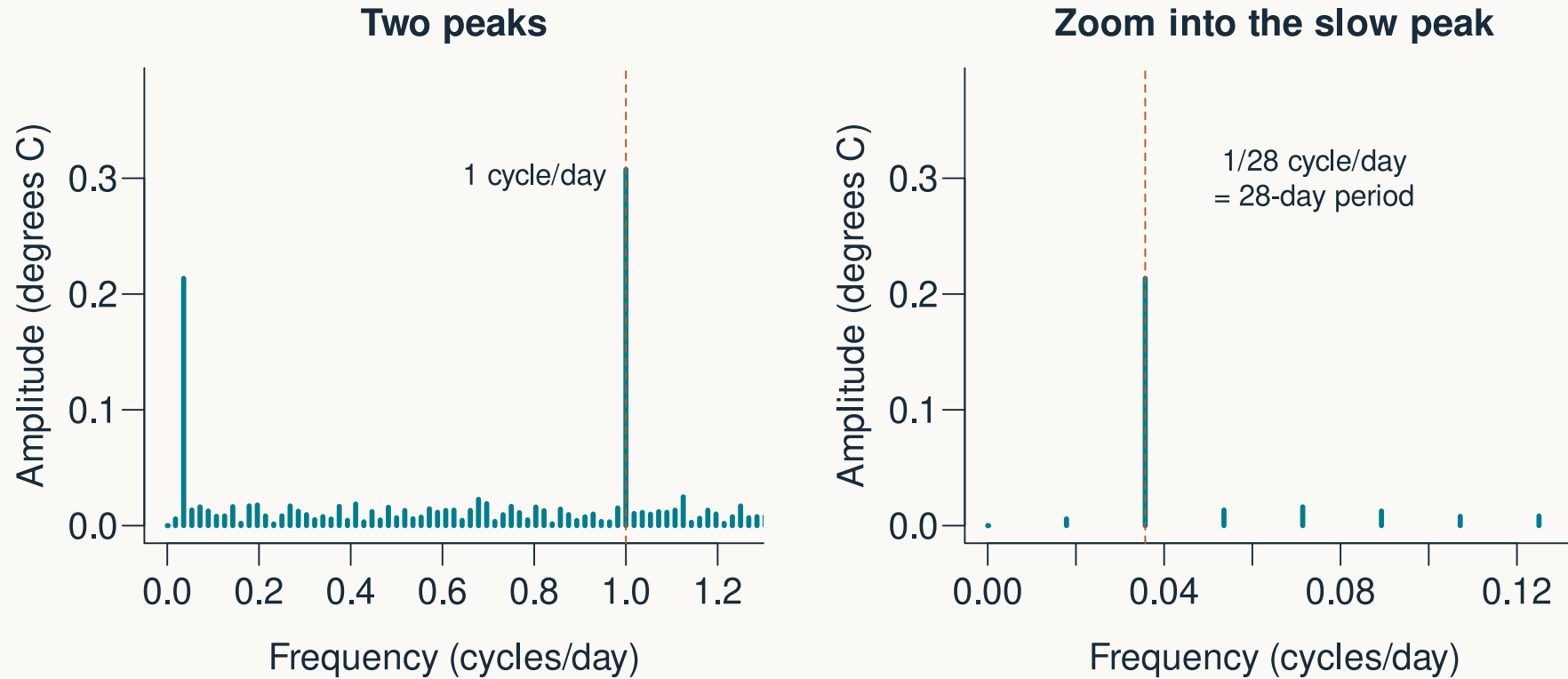
A Fourier **series** describes a periodic waveform using harmonics.

The Fourier transform changes the description

$$\underbrace{x_0, \dots, x_{N-1}}_{\text{values over time}} \xrightarrow{\text{DFT}} \underbrace{Y_0, \dots, Y_{N-1}}_{\text{frequency coefficients}}$$

Term in a paper	Meaning
DFT	Discrete Fourier transform of sampled data
FFT	A fast algorithm for calculating the DFT
Amplitude spectrum	Magnitudes plotted against frequency

Read the spectrum as a new set of axes



Horizontal: **frequency**. Vertical: **amplitude**, in the signal's units.

R: make the signal, then transform it

```
1 samples_per_day <- 24
2 N <- 56 * samples_per_day
3 t <- (0:(N - 1)) / samples_per_day
4 slow <- 0.2 * cos(2 * pi * (t - 16) / 28)
5 daily <- 0.3 * cos(2 * pi * (t - 14 / 24))
6 set.seed(10)
7 x <- 36.8 + slow + daily + rnorm(N, sd = 0.2)
8 Y <- fft(x - mean(x))
```



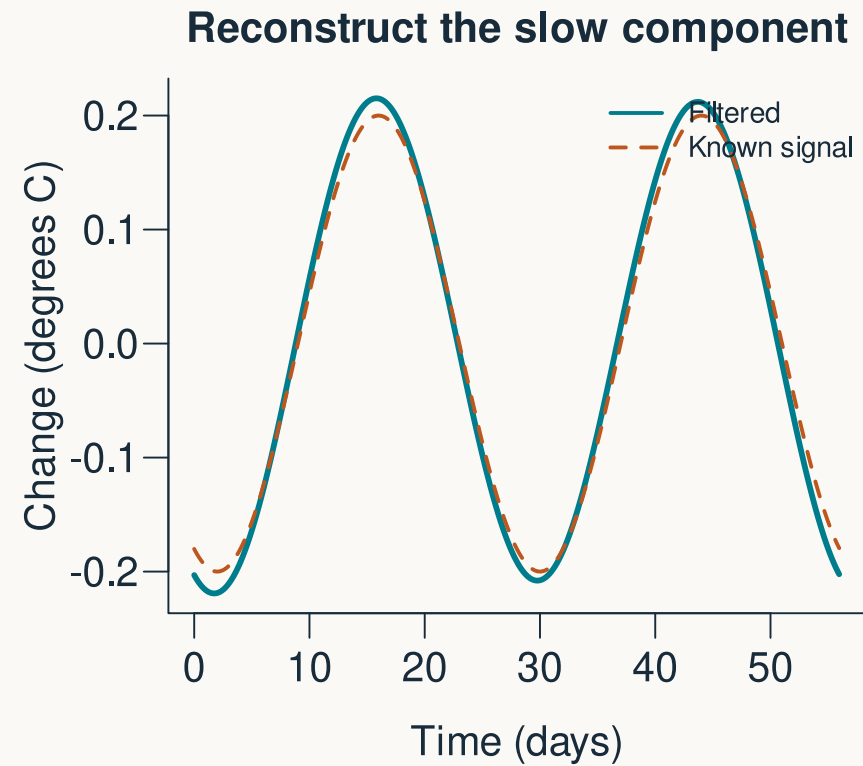
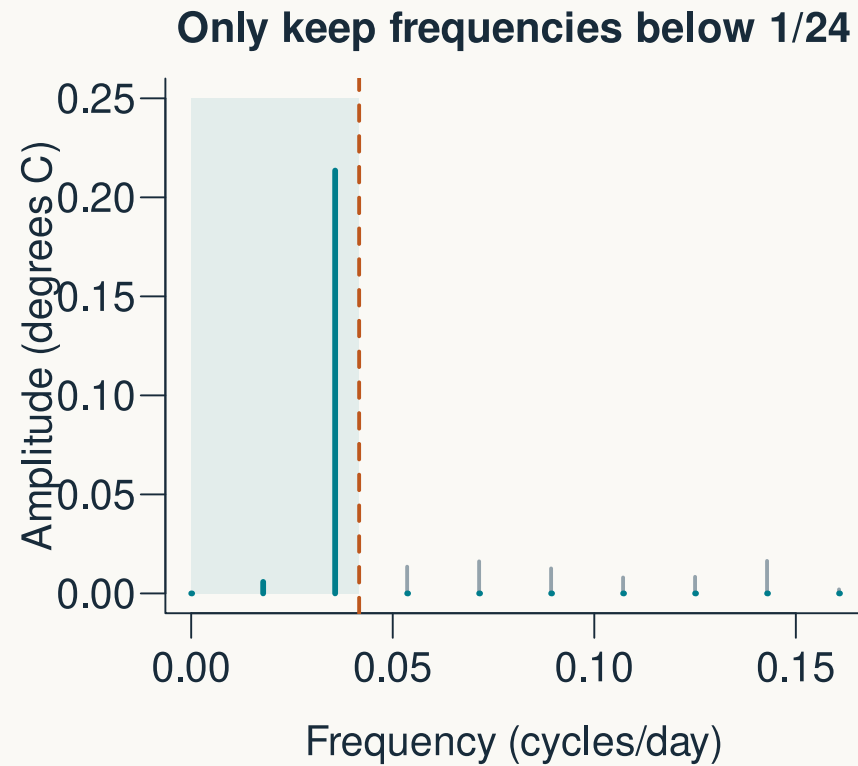
Complete, commented example: [R/fourier_transform.R](#)

R: put the frequency labels and amplitudes on the FFT

```
1 k <- 0:floor(N / 2)
2 frequency <- k * samples_per_day / N
3 amplitude <- Mod(Y[k + 1]) / N
4 paired <- k > 0 & k < N / 2
5 amplitude[paired] <- 2 * amplitude[paired]
6 plot(frequency, amplitude, type = "h", xlim = c(0, 1.3))
```

Bin spacing: $\Delta f = f_s / N = 1/56$ cycles/day.

Filtering selects which components to retain



$$|f| < 1/24 \text{ cycles/day} \iff T > 24 \text{ days} \quad (f \neq 0)$$

R: reconstruct after filtering

```
1 f_signed <- (0:(N - 1)) * samples_per_day / N
2 f_signed[f_signed > samples_per_day / 2] <-
3   f_signed[f_signed > samples_per_day / 2] - samples_per_day
4 Y[abs(f_signed) >= 1 / 24] <- 0
5 slow_change <- Re(fft(Y, inverse = TRUE)) / N
6 temperature_filtered <- mean(x) + slow_change
```

Keep both frequency partners. Divide the inverse FFT by N .

Translate a methods sentence

“We sampled at 500 Hz, differentiated position, and examined the Fourier amplitude spectrum.”

500 Hz → one sample every 2 ms.

Differentiated → estimated how quickly position changed.

Amplitude spectrum → examined the strength of each frequency component.

Three things you can now explain

See this...

Ask this...

$$y = f(x)$$

What are the inputs, parameters and units?

$$dy/dt$$

Which change per unit time is being measured?

A spectrum

What do the axes and peaks mean?

Exit check: a peak at **5 Hz** corresponds to what period?

OPTIONAL / Deriving the derivative of x^2

$$\begin{aligned}\frac{(x+h)^2 - x^2}{h} &= \frac{2xh + h^2}{h} \\ &= 2x + h \quad (h \neq 0)\end{aligned}$$

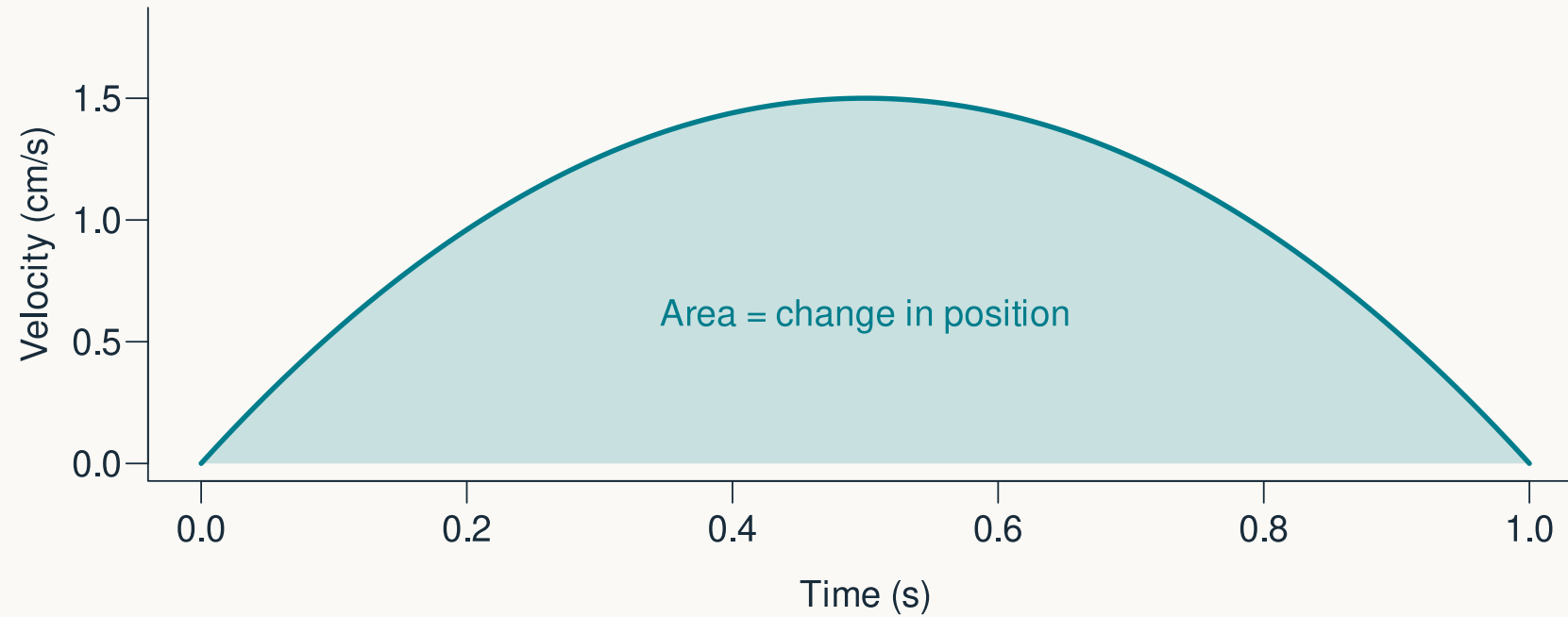
$$\lim_{h \rightarrow 0} (2x + h) = 2x$$

OPTIONAL / A small derivative reference

Function	Derivative
x^n	nx^{n-1}
e^x	e^x
$\ln x, x > 0$	$1/x$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$

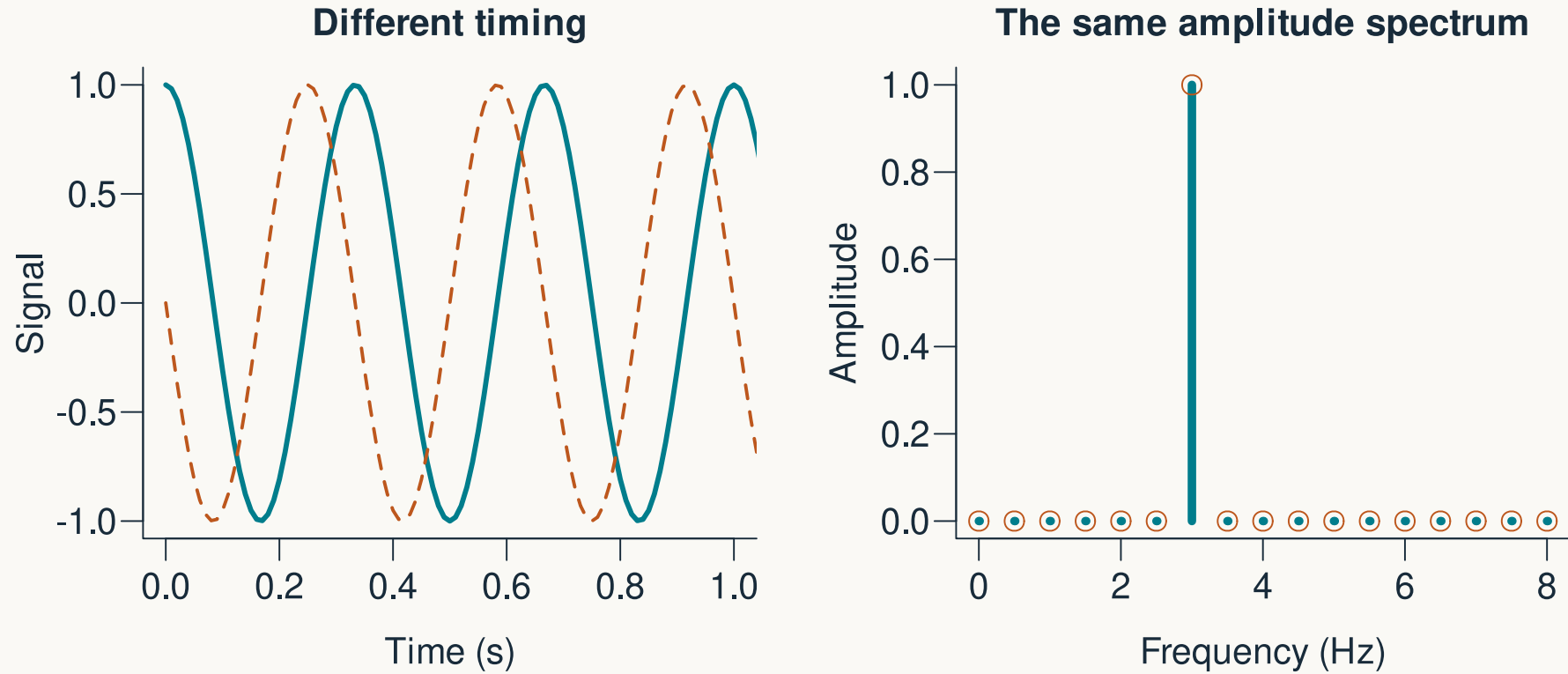
$$\frac{d}{dt}e^{-t/\tau} = -\frac{1}{\tau}e^{-t/\tau}$$

OPTIONAL / Integration adds up change



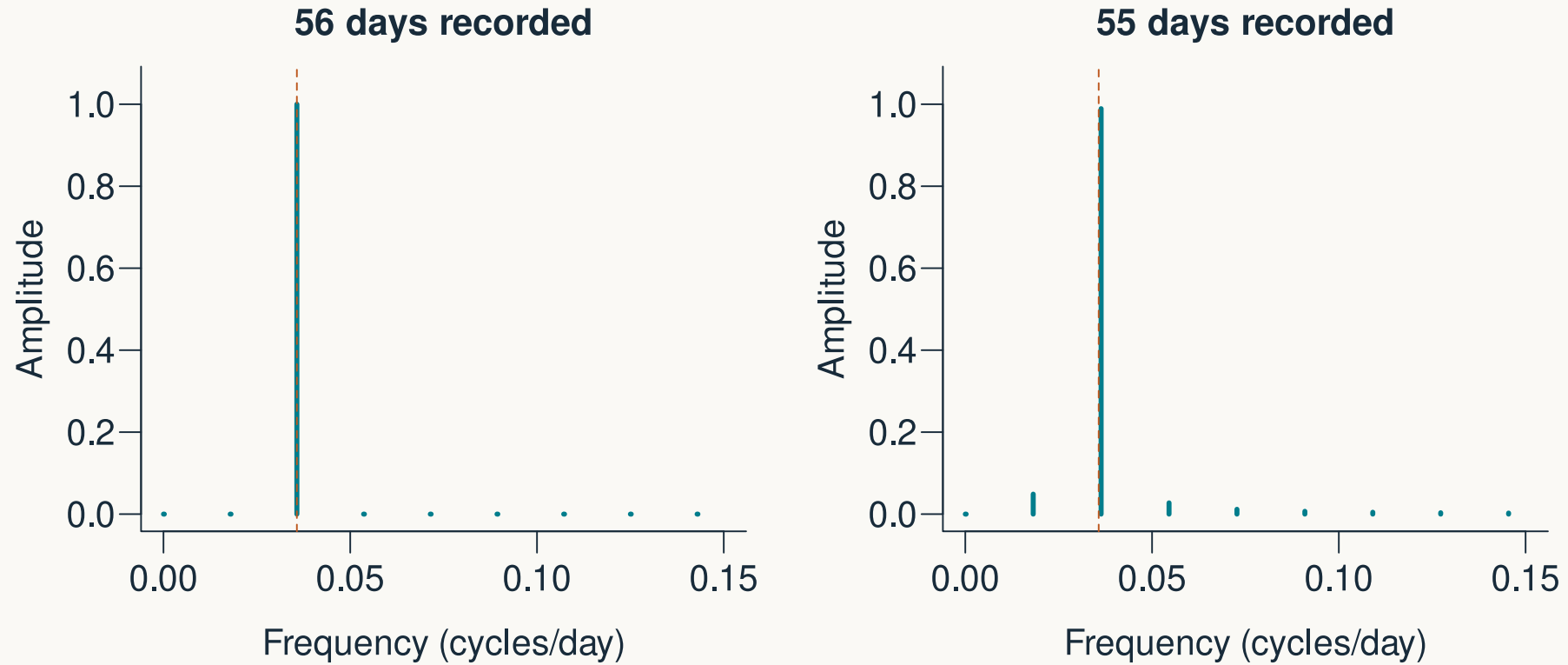
$$x(t_2) - x(t_1) = \int_{t_1}^{t_2} v(t) dt$$

OPTIONAL / An amplitude spectrum does not show timing



Reconstruction needs **phase** as well as amplitude.

OPTIONAL / Finite windows spread spectral energy



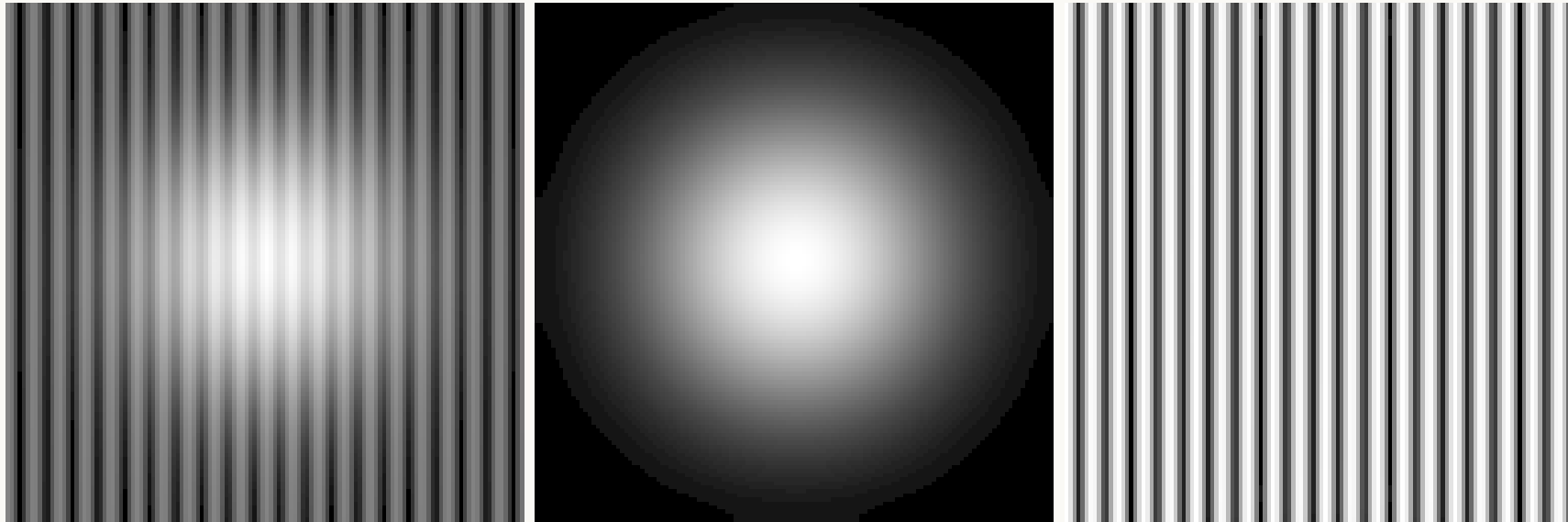
Spectral leakage depends on the recording window.

OPTIONAL / Images also have frequencies

Synthetic image

Low spatial frequencies

Remaining detail



Spatial frequency: **cycles per unit distance**, rather than per second.